International Journal of Scientific Research in Science and Technology
Available online at : www.ijsrst.com
OPEN ACCESS
Print ISSN: 2395-6011 | Online ISSN: 2395-602X
doi : https://doi.org/10.32628/IJSRST25126311

A Review on Virtual Mouse Using Hand Gestures

Prof V.W. Wankhede[1], Sakshi Wakodikar[2], Pranjali Dighore[2], Amisha Galbale[2], Akshata Nagardhane[2]

[1]Assistant professor, Department of Electronics and telecommunication, Smt. Radhikatai Pandav College of Engineering, Nagpur, Maharashtra, India

[2]B. tech Students, Department of Electronics and telecommunication, Smt. Radhikatai Pandav College of Engineering, Nagpur, Maharashtra, India

ARTICLEINFO

Article History:

Accepted : 05 Nov 2025
Published: 15 Nov 2025

Publication Issue :
Volume 12, Issue 6
November-December-2025

Page Number :
217-220

ABSTRACT

The use of hand gesture recognition in controlling virtual devices has become popular due to the advancement of artificial intelligence technology. A hand gesture-controlled virtual mouse system that utilizes AI algorithms to recognize hand gestures and translate them into mouse movements is proposed in this paper. The system is designed to provide an alternative interface for people who have difficulty using a traditional mouse or keyboard. The proposed system uses a camera to capture images of the user's hand, which are processed by an AI algorithm to recognize the gestures being made. The system is trained using a dataset of hand gestures to recognize different gestures. Once the gesture is recognized, it is translated into a corresponding mouse movement, which is then executed on the virtual screen. The system is designed to be scalable and adaptable to different types of environments and devices. All the input operations can be virtually controlled by using dynamic/static hand gestures along with a voice assistant. In our work we make use of Computer Vision algorithms to recognize hand gestures and voice commands, which work without any additional hardware requirements. The model is implemented using CNN and media pipe framework. This system has potential applications like enabling hand-free operation of devices in hazardous environments and providing an alternative interface for hardware mouse. Overall, the hand gesture-controlled virtual mouse system offers a promising approach to enhance user experience and improve accessibility through human-computer interaction.

Keyword: Computer vision, hand gesture recognition, Media-pipe, virtual mouse, PyAutoGUI, OpenCV, CNN

Copyright © 2025 The Author(s): This is an open-access article distributed under the terms of the Creative Commons Attribution **4.0 International License (CC BY-NC 4.0)**

I. INTRODUCTION

Virtual Mouse Hand Gestures technology revolutionizes computer interaction by harnessing the capabilities of MediaPipe and OpenCV, two powerful computer vision libraries. MediaPipe, developed by Google, provides accurate hand landmark detection, enabling the identification of various hand gestures in real-time. OpenCV complements MediaPipe by processing the data, refining gesture recognition through noise reduction algorithms and enhancing accuracy. The detected gestures are mapped to specific mouse actions and interpreted using mathematical algorithms.[1] Gesture recognition is a domain that involves machines using mathematical algorithms to interpret and understand human gestures. It is a subfield of computer vision. Gesture recognition involves using mathematical algorithms to interpret and understand human body movements and expressions, with a focus on facial expressions and hand gestures [2]. Computer technology has tremendously grown up over the past years and has become a necessary part of everyday life. The first part of the computer that human's access for Human Computer Interaction (HCI) is the mouse. The mouse is less appropriate for HCI in some reality things, like with Human Machine Interaction. There have been many explorations on different ways to control the computer mouse for HCI [3].

II. LITERATURE REVIEW

In this paper, an optical mouse and keyboard are created utilizing hand motions and computer vision. The computer's camera will scan the image of various hand gestures made by a user, and the computer's mouse or pointer will move in accordance with the movements of the Users can even conduct right and left clicks using various gestures. Similar to this, several gestures can be used to operate the keyboard, such as the one-finger. Without a wire or other external devices, it will function as a virtual mouse and keyboard. Gestures are a useful way to communicate, and the Covid-19 era had an impact on us. Both the keyboard and the mouse are tools used to communicate with computers. This research paper aims to use computer vision to develop an optical mouse and keyboard that can be operated.[1]

A Method for Controlling the Mouse Movement using a Real-Time Camera which enabled cursor control but provided limited functionalities. in today's world, a hand gesture recognition system is an intuitive and straightforward way for humans to communicate with machines. The use of technology-based tools, such as hand gestures, to enhance communication and interaction in education is a common occurrence in today's e-learning environment. The system utilizes MediaPipe for hand gesture recognition.[2]

There are two main approaches for Hand Gesture Recognition for HCI (Human-Computer Interaction); the first is hardware based and the second is vision-based. The engineers at Microsoft have applied for two patents detailing the Architecture to control the personal computer using hand gestures and voice input. Earlier Microsoft showed controlling devices by muscle movement, and now they are interested in gestures and voice-based systems. Their basic idea is to use voice, gesture, mouse, or keyboard as inputs to a computer.[3]

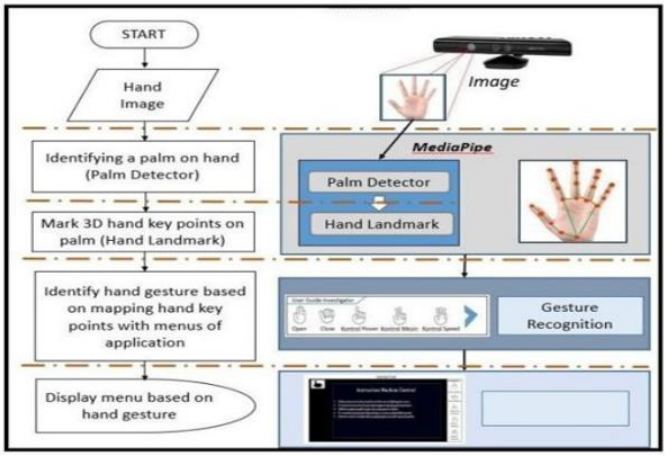
Now some others have made contributions that use Google's work with the mediapipe framework. The current gesture controlled virtual mouse uses hand gestures to perform mouse functions, in which we have control over the mouse cursor and perform certain mouse operations like left click, right click, drag and drop, volume control and brightness control etc. Efforts have been made for hand gesture recognition with camera-based detection of the hand gesture interface. In this model the hand's center is determined, and the hand's calculated radius is discovered. And using the convex hull technique, fingertip points have been determined. The hand gesture is used to control every mouse movement. The vision-based technique has been tried out in this

system. Utilized a webcam for gesture recognition and detection. And no external devices like sensors and gloves were used.[4]

Today, computer vision has advanced to the point where a machine can recognize its owner by running a straight forward picture processing programmer. People use this vision in many parts of daily life at this stage of evolution, including face recognition, colour detection, automatic cars, etc. In this project, an optical mouse and keyboard are created utilising hand motions and computer vision. The computer's camera will scan the image of various hand gestures made by a user, and the computer's mouse or pointer will move in accordance with the movement of the movements.[5]

III.GENERAL METHODOLOGY

3.1 GENERAL BLOCK DIAGRAM



3.2 SOFTWARE USED

Open CV

OpenCV performs a multifaceted position withinside the project, basically specializing in digital digicam configuration, body acquisition, and preprocessing tasks. It allows the initialization and configuration of the internet digital digicam for video capture, making sure a consistent flow of frames for analysis. Moreover, OpenCV plays crucial preprocessing tasks, consisting of satiation area conversion and noise reduction, which decorate the nice and reliability of hand gesture detection and popularity. OpenCV is an open-source library for computer vision that can be used on various platforms. Computer vision is a field of study that focuses on enabling computers to understand and interpret visual information from digital images or videos. OpenCV is designed to be a comprehensive solution for image processing and analysis, including tasks such as face detection, object detection, and image segmentation. It provides a wide range of tools and functions that make it easier to implement these tasks, even for those without a deep understanding of computer vision techniques. Regarding the proposed system, access to it is restricted to registered users only. This means that users must first register for the system and create a unique username and password. They can then log in using the same credentials each time they want to use the system. This helps to ensure that only authorized users can access the system and perform the intended tasks and provides a level of security and privacy for user data.

Media Pipe

In the project, Media Pipe serves as an essential issue for real-time hand landmark detection and gesture popularity. Media Pipe presents superior algorithms and fashions for appropriately figuring out key factors at the hand, allowing gesture popularity in the digital interface. By leveraging Media Pipe's capabilities, the machine can interpret complicated hand moves and translate them into corresponding actions, which include controlling the digital mouse cursor, executing keyboard inputs, or adjusting the extent level. Media Pipe framework is an open-source framework by Google used for applying with the machine learning pipeline. It is a cross-platform development tool, and it is constructed using time series data. This framework is commonly used with various audio and video applications. Moreover, it is used for constructing and analyzing the system using graphs and system development for application purposes. Developers can quickly construct intricate pipelines using Media Pipe that combine numerous algorithms and processes and execute in real-time on a variety of h/w platforms, like CPUs, GPUs, and

specialized accelerators like Google's Edge TPU. Additionally, it supports several programming languages, like C++, Python, and Java.

PyAutoGUI

PyAutoGUI serves as a vital factor for simulating person interactions primarily based totally on detected hand gestures. PyAutoGUI permits the machine to simulate mouse movements, keyboard inputs, and extent changes in reaction to identified gestures, efficiently bridging the distance among gesture reputation and machine moves. By translating detected gestures into unique moves in the digital interface, PyAutoGUI guarantees seamless and intuitive interaction between people, improving the general usability and accessibility of the machine.

IV. CONCLUSION

It can be concluded that virtual mouse systems using hand gestures are becoming an advanced and effective method for human–computer interaction. All studies show that with the help of computer vision and artificial intelligence, hand movements can be accurately detected to perform tasks like cursor movement, clicking, scrolling, and even typing. These systems provide a touchless, hygienic, and user-friendly experience compared to traditional mouse and keyboard devices. It highlights that gesture-based virtual mouse technology is simple to use, cost-efficient, and can work on basic hardware with a normal camera. This makes the system suitable for daily use as well as for advanced applications in education, automation, smart devices, and assistive tools. Overall, the research strongly supports that hand-gesture virtual mouse systems have a promising future and can play an important role in modern computing.

REFERENCES

[1]. Khushi Pathak, Aniket Joshi, Aryan Kalra, Shraddha Patil, Rajendra Babar, “Virtual Mouse and Keyboard with Hand Gestures”, International Journal of Creative Research Thoughts (IJCRT) ISSN: 2320-2882; Volume: 12, Issue 3 March 2024.

[2]. Karan Kharbanda, Utsav Sachdeva, “Gesture Controlled Virtual Mouse Using Artificial Intelligence”, International Research Journal of Modernization in Engineering Technology and Science (IRJETS) ISSN:2582-5208; Volume:05, Issue 01 January 2023.

[3]. Yash Yadav, Abhishek Hazare, Rushab Dungarwal, Pratham Sharma, Shefali Raina, “Virtual Mouse Control Using Hand Gestures”, International Research Journal of Engineering and Technology (IRJET) ISSN:2395-0056, ISSN:2395-0072; Volume:10, Issue 01 January 2023.

[4]. Kavitha R, Janasruthi S U, Lokitha S, Tharani G, “Hand Gesture Controlled Virtual Mouse Using Artificial Intelligence”, IJARIIE, ISSN:2395-4396; Volume:09, Issue 2 2023.

[5]. Pratiksha Kadam, Prof. Minal Junagre, Sakshi Khalate, Vaishnavi Jadhav, Pragati Shewale, “Gesture Recognition based Virtual Mouse and Keyboard “, International Journal for Research in Applied Science & Engineering Technology (IJRASET) ISSN: 2321-9653; Volume 11 Issue V May 2023.